CSC 370: Databases Third Normal Form

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Which attribute sets are keys?

R(A,B,C,D)

$A \rightarrow BC$ $AC \rightarrow D$ $C \rightarrow A$

- ❏ {A}
- ❏ {A, C}
- ❏ {C}
- ❏ {A, B, C, D}

Today's Goal:

You can decompose a relation into 3NF after identifying conditions that make 3NF an appropriate alternative.

Outline

1. Review minimal bases
2. Work through an example in which BCNF decomposition leads to a loss of FDs
3. Prime attributes
4. Third Normal Form (3NF)
5. Summary / Exercise

Last class...

Closure of a set of FD's

The closure of a *set of functional dependencies* (as opposed to a set of attributes) is:

- The closure of all possible subsets of the attributes

For example, given:
R(A,B,C)
$A \rightarrow B$
$C \rightarrow B$

The closure for $\{A \rightarrow B, C \rightarrow B\}$ is:

$\{\}^+ = \{\}$
$\{A\}^+ = \{A,B\}$
$\{B\}^+ = \{B\}$
$\{C\}^+ = \{B,C\}$
$\{A,B\}^+ = \{A,B\}$
$\{A,C\}^+ = \{A,B,C\}$
$\{B,C\}^+ = \{B,C\}$
$\{A,B,C\}^+ = \{A,B,C\}$

Source: Garcia-Molina et al. (2007), §3.2.7

Basis of a set of FD's

We can then say that two *sets of functional dependencies* are equivalent iff:

- They have the same *closing set*
- When two sets of FD's are equivalent, we say they are a *basis* for each other

Given:
R(A,B,C)
$A \rightarrow B$
$C \rightarrow B$

What other sets of FD's form a basis for this set ($\{A \rightarrow B, C \rightarrow B\}$)

Some examples:

$A \rightarrow AB$	$A \rightarrow B$	$AB \rightarrow B$
$C \rightarrow BC$	$B \rightarrow B$	$A \rightarrow AB$
	$C \rightarrow B$	$C \rightarrow B$

If it's clear that two sets of FD's can be equivalent, then we can ask if any of them are better than any others.

Source: Garcia-Molina et al. (2007), §3.2.7

Minimal Basis

Given a set of FD's *F*, we say that they form *a minimal basis* iff:

- *Every FD has a singleton right-hand side*
- *If any FD is removed, we no longer have a basis*
- *If we remove any attribute from the left-hand side of any FD, we no longer have a basis*

In other words, there is no simpler set of bases that form a basis for *F*.

Are both of these minimal bases?

F	**F'**
$A \rightarrow B$	$A \rightarrow D$
$B \rightarrow D$	$D \rightarrow B$
$D \rightarrow A$	$B \rightarrow A$

Source: Garcia-Molina et al. (2007), §3.2.7

Projecting Functional Dependencies

Let's project R(A,B,C,D,E) onto S(A,B,C) and then project the following FD's onto S as well:

$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

1. *Remove all triviality and semi-triviality*
2. *Remove everything not in S*

$A \rightarrow ABD$
~~$B \rightarrow BD$~~
~~$AB \rightarrow ABD$~~
$AC \rightarrow ABCDE$
~~$AD \rightarrow ABD$~~
~~$AE \rightarrow ABDE$~~
~~$BC \rightarrow BCD$~~
~~$BD \rightarrow BD$~~
~~$BE \rightarrow BDE$~~

~~$CD \rightarrow CDE$~~
~~$CE \rightarrow CE$~~
~~$DE \rightarrow DE$~~
~~$ABC \rightarrow ABCDE$~~
~~$ABD \rightarrow ABD$~~
~~$ABE \rightarrow ABDE$~~
~~$ACD \rightarrow ABCDE$~~
~~$ACE \rightarrow ABCDE$~~
~~$ADE \rightarrow ABDE$~~

~~$BCD \rightarrow BCDE$~~
~~$BCE \rightarrow BCDE$~~
~~$BDE \rightarrow BDE$~~
~~$CDE \rightarrow CDE$~~
~~$ABCD \rightarrow ABCDE$~~
~~$ABCE \rightarrow ABCDE$~~
~~$ABDE \rightarrow ABDE$~~
~~$ACDE \rightarrow ABCDE$~~
~~$BCDE \rightarrow BCDE$~~

Source: Garcia-Molina et al. (2007), §3.2.8

Projecting Functional Dependencies

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$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

1. *Remove all triviality and semi-triviality*
2. *Remove everything not in S*
3. *Reduce to a minimal basis*

$A \rightarrow B$
$AC \rightarrow B$

- We can remove the second FD and still have a basis.
  Therefore the FD's for S are:

F'
$A \rightarrow B$

Source: Garcia-Molina et al. (2007), §3.2.8

Take-Home Exercise: Example 3

What is the set of FD's from projecting MovieShowings(title, theatre, city) onto Theatres(theatre,city), given the set of FD's below?

theatre → city
title city → theatre

We will revisit this example in the 3NF lesson.

Source: Garcia-Molina et al. (2007), §3.4.4

Dependency preservation

Dependency Preservation

Let's decompose into BCNF the following schema and FD's and rigorously/corrected project FDs:

CinemaShowings(city, theatre, title)
theatre → city
city, title → theatre

This scenario ensures that two theatres in the same city don't compete for viewers of the same movie

Source: Garcia-Molina et al. (2007), §3.4.4

Post-mortem

Let's decompose into BCNF the following schema and FD's:

CinemaShowings(city, theatre, title)
theatre → city
city, title → theatre

Showings(theatre, title)
Theatres(theatre, city)

Why did we break apart the key {city, title}?

- Only half of that key appeared in the closure of the violating FD!
- Where did the FD {city, title} → {theatre} go?
- Our BCNF database doesn't enforce the second constraint with key constraints anymore!
    - So much for opportunities to monopolise competition and engage in anti-competitive practices! 😭 (It's the textbook's example, not mine.)

Source: Garcia-Molina et al. (2007), §3.4.4

Prime attributes

Prime Attributes

We can recognise the problem from the previous example by identifying *prime* attributes:

- An attribute is *prime* iff it is part of a key
- **Note**: that is a *key*, not a *superkey*!

Source: Garcia-Molina et al. (2007), §3.4.4

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CinemaShowings(city, theatre, title)
theatre → city
city, title → theatre

Observe that {theatre, title} and {city, title} are both keys. Thus:

- *city* is prime
- *theatre* is prime
- *title* is prime

Source: Garcia-Molina et al. (2007), §3.4.4

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BCNF decomposition will usually lose constraints if some FD has an antecedent involving a prime attribute that is not a key.

Source: Garcia-Molina et al. (2007), §3.4.4

Prime Attributes: Example 2

- Which are the prime attributes in the relation below?

R(A, B, C, D, E)
AB → C
BC → D

Prime Attributes: Example 2

- Which are the prime attributes in the relation below?

R(A, B, C, D, E)
AB → C
BC → D

Observe that {A,B, E} was the only key. Thus:

- *A* is prime
- *B* is prime
- *C* is **not** prime
- *D* is **not** prime
- *E* is prime

Which attributes are prime?

R(A,B,C,D)

$A \rightarrow BC$ $AC \rightarrow D$ $C \rightarrow A$

- ❏ A
- ❏ B
- ❏ C
- ❏ D

3NF

Third Normal Form (3NF)

A relation R over attributes C with functional dependencies F is in 3NF iff:

Every FD f in F is in 3NF.

An FD f:X→Y is in 3NF iff:

X is a superkey
OR
Every attribute in Y is "prime"

Equivalently:

f is not a BCNF violation
OR
Every attribute in Y is "prime"

Source: Garcia-Molina et al. (2007), §3.5.1

Revisiting the movie example

Let's decompose into **3NF** the following schema and FD's:

R(city, theatre, title)
f_0: theatre → city
f_1: city, title → theatre

Observe that:
$\{city,title\}^+ = \{city, title, theatre\}$
$\{theatre,title\}^+ = \{city, title, theatre\}$

f_1 is not a BCNF violation because its antecedent is a superkey. Therefore it is also in 3NF
f_0 *is* a BCNF violation because $\{theatre\}+ \neq \{city, theatre, title\}$;
- However, city is *prime* (part of key {city,title}), so f_0 is in 3NF

Because this relation is already in 3NF, we do not decompose it and the dependencies are preserved.

Source: Garcia-Molina et al. (2007), §3.5.1

3NF Synthesis Algorithm

Input: A relation R with attribute set C and a set of functional dependencies F
Output: A set of projections of R and F that are in 3NF, have a lossless join, and preserve dependencies

1. Let G be a minimal basis for F
2. For each functional dependency g: $X \rightarrow Y$ in G:
   a. Create a sub-relation by projecting R onto $X \cup Y$
3. If no sub-relation has a superkey for R:
   a. Create a sub-relation with *some* key
4. Return all sub-relations

Source: Garcia-Molina et al. (2007), §3.5.2

3NF Synthesis Algorithm Example

Consider R(title,city,theatre,genre,start_time) with functional dependencies
{city, title} → {theatre}
{theatre} → {city}
{title} → {genre}

Would the BCNF decomposition work on this example?
Let's work through the 3NF synthesis algorithm together…

Source: Garcia-Molina et al. (2007), §3.5.2

Summary / Exercise

Summary

- When a schema is not normalised, anomalies (i.e., data inconsistencies and constraint violations) are likely to occur
- There are a number of different normal forms to reduce/prevent these anomalies, each with different advantages
  - 3NF preserves all the functional dependencies that we have identified
  - BCNF eliminates redundancy due to FD's
- Both decomposition algorithms are *non-deterministic*
- After normalisation, we have completed the logical design

Source: Garcia-Molina et al. (2007), §3.6.6

Take-home Exercise

Consider R(A,B,C,D) with functional dependencies
$A \rightarrow BC$
$B \rightarrow A$
$AB \rightarrow C$

Source: Garcia-Molina et al. (2007), §3.5.2